

CSET485 – AI and Society
Assignment #1
10 Marks

Objective: Build a predictive model on your assigned dataset, deliberately stress-test it using statistical and adversarial techniques and defend your findings in writing using your own computed numbers. This assignment examines how predictive models used in high-stakes societal decisions, such as criminal risk scoring, credit approval, or income classification, can appear statistically sound while still concealing bias, hidden confounds, and fairness trade-offs that only surface through rigorous, hands-on scrutiny.

Dataset: You will work with one of three real-world datasets, assigned based on the last digit of your roll number:

Last Digit	Assigned Dataset	Dataset Link
0, 1, 2	COMPAS Recidivism Risk Score Dataset	https://github.com/propublica/compas-analysis/raw/master/compas-scores-two-years.csv
3, 4, 5	German Credit Data	https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/
6, 7, 8, 9	Adult Census Income Dataset	https://archive.ics.uci.edu/ml/machine-learning-databases/adult/

Additionally, use the last 3 digits of your roll number as the random seed (random_state) for your train-test split (70% train, 30% test).

Data Preprocessing: Before any analysis, document your handling of missing values, categorical encoding, feature scaling, and outliers. This documentation must be included in your submission.

Part 1 — Build

- **Fit both models on the same dataset:**

- **Model A (Linear Regression):**
 - COMPAS students: predict `decile_score` (continuous risk score, 1–10)
 - German Credit students: encode approval as 1=approve, 0=reject; predict the probability using OLS
 - Adult Income students: encode income as 1 if >\$50K, 0 if ≤\$50K; predict the probability using OLS

- **Model B (Logistic Regression):**

- COMPAS students: predict `is_recidivist` (binary, 1=recidivated, 0=no recidivism)
- German Credit students: predict approval (1=approve, 0=reject)
- Adult Income students: predict income >\$50K (1=yes, 0=no)

- **Compare on test set (30% of data):**

- Take Model A's predictions and threshold at the **median** predicted value → classify as "high risk" if $\geq$ median, else "low risk"
 - Take Model B's predicted probabilities and classify as "high risk" if ≥ 0.5 , else "low risk"
 - **Report:**
 - Confusion matrix showing agreement/disagreement between the two thresholding approaches
 - Count and percentage of mismatched cases (how many individuals flagged differently?)
 - **Explain:** Why do these two approaches produce different classifications?
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Part 2 — The Trap

2.1: Simpson's Paradox Hunt

Find a subgroup in your dataset where the overall trend and the subgroup trend move in opposite directions.

- Split the data by one demographic/social variable (e.g., age, gender, race). Fit logistic regression on: (1) full dataset, (2) subgroup A, (3) subgroup B.
- **Report:** Regression coefficients for at least one key predictor across all three models. Do the coefficients reverse sign or become much weaker/stronger across subgroups?
- **If no reversal exists:** Identify a subgroup where the effect is significantly stronger or weaker (use confidence intervals or p-values to show the difference). Document this instead.
- **Explain:** Using your actual coefficients, why does the trend reverse (or differ significantly)?
- **Policy implication:** How would policy recommendations change if subgroup analysis were ignored?

2.2: Omitted-Variable Bias

Deliberately drop a variable you know is likely confounding the relationship.

- Fit two logistic regression models: (1) Full model (all features), (2) Reduced model (without the chosen confounding variable)

- **Report:** Before-and-after coefficients for at least 4 predictors. Calculate the percentage change for each.
- **Which coefficient shifted the most?** What does this reveal about trusting a model without stress-testing for confounds?
- **Reflect:** In the real world, how would you detect if your current model has a similar hidden confound that you cannot see?

2.3: Adversarial Subset Construction

Identify cases where your logistic model makes confident, incorrect predictions.

- On your test set, find instances where your model predicts $P(\text{outcome}=1) \geq 0.75$ (high confidence) but the actual outcome is 0 (false positive), OR predicts $P \leq 0.25$ (high confidence negative) but actual outcome is 1 (false negative).
- Extract approximately 50 such rows (if fewer exist, use all; if more, randomly sample 50).
- **Report:**
 - Total count and percentage of these high-confidence wrong predictions
 - Demographics of the 50 rows (e.g., age distribution, gender breakdown) compared to full test set
 - What common characteristic do these 50 rows share? (e.g., "94% have prior_crimes ≥ 5 ")
- **Explain:** Statistically, why does your model confidently fail on this subset? (Hint: missing interaction term? data distribution shift? non-linear relationship ignored?)

2.4: Fairness Trade-Off Analysis

Calculate three fairness metrics across a range of decision thresholds to reveal unavoidable trade-offs.

Methodology:

- Choose one sensitive variable (e.g., race, gender, age groups). Split data into Group A and Group B.
- Sweep your logistic regression's decision threshold from 0.05 to 0.95 in steps of 0.1 (i.e., $t \in \{0.05, 0.15, 0.25, \dots, 0.95\}$).
- For each threshold t , classify as 1 if $P \geq t$, else 0.
- For each threshold, compute:
 - **Demographic Parity (DP):** $P(\hat{Y}=1|\text{Group A}) / P(\hat{Y}=1|\text{Group B})$ — ideal value is 1.0
 - **Equalized Odds (EO):** $\min(\text{TPR}_A/\text{TPR}_B, \text{FPR}_A/\text{FPR}_B)$ — ideal value is 1.0
 - **Predictive Parity (PP):** $P(Y=1|\hat{Y}=1, \text{Group A}) / P(Y=1|\hat{Y}=1, \text{Group B})$ — ideal value is 1.0

Report:

- Table showing DP, EO, PP for each threshold value

- **Trade-off chart:** X-axis = Demographic Parity (0–1.5), Y-axis = Equalized Odds (0–1.5), with each point labeled by threshold value
 - **Quantify the trade-off:** "If you improve DP from X to Y (a Z% gain), EO drops from A to B (a W% loss). The trade-off cost is W/Z."
 - **Key insight:** No single threshold satisfies all three metrics. Which trade-off would you recommend, and why?
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Part 3 — Descriptive Analysis Questions (around 250–300 words each)

Each answer must directly reference specific numbers, coefficients, or results from your own analysis. A generic answer with no numbers attached will score zero on that question, regardless of how well-written it is.

1. Explain in detail why the threshold-based approach and the direct logistic regression approach did, or did not, flag the same individuals as "high risk" in your data. Quote the specific count/percentage of mismatched cases. Reference the median predicted value from linear regression and how it compares to the 0.5 threshold in logistic regression.
2. Describe the subgroup you found (or the subgroup with significantly different effects). Using your actual regression coefficients from the full model, subgroup A, and subgroup B, explain why the trend reverses or differs. Cite the exact coefficient values and confidence intervals. Explain what policy conclusion would change if the subgroup were ignored.
3. State which variable you dropped for the omitted-variable test. Report the specific before/after shift in coefficients for at least 4 predictors—include a table. Calculate the percentage change for each. Which coefficient shifted the most? What does this reveal about how much you can trust a model you did not stress-test this way?
4. As a practitioner, how would you detect a hidden confound in your model that you cannot currently see? Propose one concrete, real-world method (e.g., domain expert audit, external validation on a different dataset, residual analysis, instrumental variable approach). This question has no clean answer; you will be graded on the realism and concreteness of your proposal.
5. Describe your 50-row adversarial subset. What common characteristic do these rows share, and what percentage of the subset exhibits this trait? Statistically, why does this characteristic cause your model to fail confidently? Reference your model's coefficients or learned decision boundaries to explain the failure mechanism.
6. Using your own fairness trade-off chart from Part 2.4, explain in concrete terms what the trade-off cost means for a real person affected by this decision. Describe a specific, realistic scenario: two individuals (one from each demographic group) and their outcomes under different decision thresholds. Show how prioritizing one fairness metric harms another group.
7. Choose a specific decision threshold from your sweep (e.g., $t=0.35$). Argue, using at least 3 specific numbers from your analysis (DP, EO, PP, accuracy, subgroup-specific rates), why this threshold is defensible from a fairness standpoint. Describe the strongest counter-argument a critic could raise against your choice and acknowledge its validity.

8. If a policymaker had to choose between prioritizing demographic parity or equalized odds for your dataset's specific context, which would you recommend, and why? Cite your own trade-off numbers from Part 2.4. Acknowledge what fairness metric you are sacrificing and why that is acceptable in your view.
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Part 4 — Reflection (around 500 words)

"If a policy-maker asked you: 'just give me one number: should we deploy this model or not?', what would you say, and what is the limitation of that 'one number' answer that you discovered yourself through your analysis?"

Structure your answer as follows:

1. **Direct answer** (1–2 sentences): State clearly whether you would recommend deployment, conditional deployment, or rejection, with one headline reason.
 2. **Why a single number fails** (300–400 words): Identify at least 3 limitations of a single-number answer using evidence from your analysis:
 - E.g., overall accuracy hides subgroup performance gaps
 - E.g., threshold choice is a values question, not a data-driven fact
 - E.g., fairness metrics conflict (improving one harms another)
 - E.g., model confidently fails on predictable subsets
 - E.g., hidden confounds remain unknowable
 3. **Grounding in evidence** (50–100 words): Cite specific numbers from Parts 1–2 to support each limitation.
 4. **Closing** (50 words): Reflect on the meta-lesson: responsible AI deployment requires transparency about trade-offs, not a single metric.
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Submission Requirements: Submit a single PDF report containing your data preprocessing section (describing your handling of missing values, categorical encoding, feature scaling, and outliers), Part 1 (model comparison and mismatch analysis), Part 2 (all four stress-tests with tables and charts), Part 3 (eight descriptive analysis answers of ~250–300 words each), and Part 4 (reflection essay of ~500 words).

Additionally, submit your fully reproducible code as a Jupyter notebook via a GitHub link with clear documentation of all computations; every number and result cited in your PDF must be regenerable by running your code from start to finish.

Submission Deadline - 19 Oct 2026, 23:59 IST